

1.3.1 Example:

$$1 + \dots + \frac{1}{9} + \frac{1}{10} + \frac{1}{11} + \frac{1}{12} + \frac{1}{13} + \frac{1}{14} + \dots \quad \mathcal{D}$$

Do these series converge or diverge?

$$\left. \begin{array}{l} \sum_{n=1}^{\infty} \frac{3}{n^2} = 3 \sum_{n=1}^{\infty} \frac{1}{n^2} \\ \mathcal{C} \\ p=2 \end{array} \right| \left. \begin{array}{l} \sum_{n=1}^{\infty} \frac{n^3+n}{n^4} \quad \mathcal{D} \\ \sum_{n=1}^{\infty} \frac{n^3}{n^4} + \sum_{n=1}^{\infty} \frac{n}{n^4} \\ \sum_{n=1}^{\infty} \frac{1}{n} + \sum_{n=1}^{\infty} \frac{1}{n^3} \\ \mathcal{D} \quad \mathcal{C} \end{array} \right| \left. \begin{array}{l} \sum_{n=1}^{\infty} \frac{1}{n+10} \quad \mathcal{D} \\ \text{It's the harmonic series with the first 10 terms missing} \end{array} \right| \left. \begin{array}{l} \sum_{n=1}^{\infty} \frac{1}{n^3+n} < \sum_{n=1}^{\infty} \frac{1}{n^3} \\ \mathcal{C} \end{array} \right| \boxed{\sum \frac{1}{n^3+n} < \sum \frac{1}{n^3} = \text{Finite}}$$

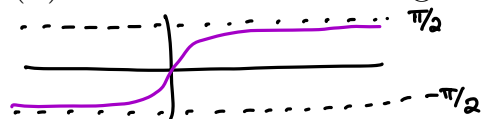
2 The Integral Test

Integral Test

Let $f(x)$ be a continuous function that decreases to 0 as $x \rightarrow \infty$, and
Let $a_n = f(n)$. Then

$$\sum_{n=1}^{\infty} a_n \text{ is convergent iff } \int_1^{\infty} f(x) dx \text{ is convergent.}$$

Note: This test says that $\sum_{n=1}^{\infty} a_n$ and $\int_1^{\infty} f(x) dx$ do the same thing. It does NOT say that they are equal.



2.0.1 Example:

$$\sum_{n=1}^{\infty} \frac{1}{n^2+1} \quad \int_1^{\infty} \frac{1}{x^2+1} dx = \arctan x \Big|_1^{\infty} = \arctan(\infty) - \arctan(1) \\ = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \quad \leftarrow \text{The integral converges to } \frac{\pi}{4}.$$

The series also converges.

2.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{\ln n}{n} \quad \int_1^{\infty} \frac{\ln x}{x} dx = \int_0^{\infty} u du = \frac{1}{2} u^2 \Big|_0^{\infty} = \frac{1}{2} (\infty)^2 - \frac{1}{2} (0) \\ = \infty - 0 = \infty$$

$$u = \ln x \quad \text{Top: } u = \ln(\infty) = \infty \\ du = \frac{1}{x} dx \quad \text{bot: } u = \ln 1 = 0$$

The integral diverges.

The series also diverges.

11.4 Comparison Tests

So far, we know about

Geometric Series $\sum_{n=1}^{\infty} ar^{n-1}$

- converges if $|r| < 1$; diverges if $|r| \geq 1$
- If it converges, the sum is $\frac{a}{1-r}$

p -Series $\sum_{n=1}^{\infty} \frac{1}{n^p}$

- Converges if $p > 1$, diverges if $p \leq 1$
- No simple formula for the sum.

0.0.1 Example:

$$\sum_{n=1}^{\infty} \frac{1}{3^n + n^2}$$

neither

Compare with $\sum_{n=1}^{\infty} \frac{1}{3^n} \leftarrow \text{Converges.}$

$$\frac{1}{3^n + n^2} < \frac{1}{3^n}$$

↑
bigger denominator,
smaller #.

$$\sum_{n=1}^{\infty} \frac{1}{3^n + n^2} < \sum_{n=1}^{\infty} \frac{1}{3^n} = \frac{\frac{1}{3}}{1 - \frac{1}{3}} = \frac{1}{2}$$

↑
this must also
converge (to
some sum that's
less than $\frac{1}{2}$).

$\frac{\frac{1}{3}}{\frac{2}{3}}$

0.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{3^n + n^2}{2^n}$$

compare with $\sum \frac{3^n}{2^n}$, which diverges

$$\frac{3^n + n^2}{2^n} > \frac{3^n}{2^n}$$

↑
Bigger numerator.
∴ Bigger #.

$$\text{so } \sum \frac{3^n + n^2}{2^n} > \sum \frac{3^n}{2^n} = \infty$$

↑
This also diverges to ∞
(faster than the other series).

1 Brief Review of comparing Fractions

Comparing Fractions

Suppose we have the fraction

$$F = \frac{\text{numerator}}{\text{denominator}},$$

and that both numerator and denominator are positive.

Facts:

- If numerator grows larger, F gets larger (i.e. approaches ∞).
- If denominator grows larger, F gets smaller (i.e. approaches 0).
- If **both** grow larger, we can't (without more info) say whether F gets larger or smaller.

1.0.1 Example:

$$\frac{1}{2} > \frac{1}{10}$$

$$\frac{37}{25} < \frac{40}{25}$$

$$\frac{3}{1232} < \frac{3}{1100}$$

2 The (regular) Comparison Test

The Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

- if $\sum a_n$ converges and $b_n \leq a_n$ for all $n \geq 1$, then $\sum b_n$ also converges.
- if $\sum a_n$ diverges and $b_n \geq a_n$ for all $n \geq 1$, then $\sum b_n$ also diverges.

2.0.1 Example:

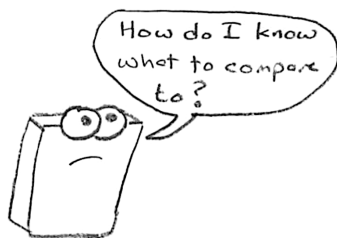
$$\sum_{n=1}^{\infty} \frac{2}{n^3 + 4}$$

Compare with $\sum \frac{2}{n^3}$, which converges.

$$\sum \frac{2}{n^3 + 4} < \sum \frac{2}{n^3} \text{ converges}$$

also converges
↑
bigger denom.
smaller number

converges by comparison
with $\sum \frac{2}{n^3}$.



A: As a rule of thumb, compare with the series you get by stripping off everything except the dominant terms on top and bottom. If the dominant term is a sine or cosine, replace it with 1; if it's arctan, replace it with $\frac{\pi}{2}$.

2.0.2 Example:

$$\sum_{n=1}^{\infty} \frac{4}{n^2 + 12n}$$

compare with $\sum \frac{4}{n^2}$, which converges

also converges. $\rightarrow \sum \frac{4}{n^2 + 12n} < \sum \frac{4}{n^2}$

bigger denom
smaller number

2.0.3 Example:

$$\sum_{n=1}^{\infty} \frac{3n^2 + 2n - 1}{n^2 + 5}$$

~~Compare with $\sum \frac{3n^2}{n^2} = \sum 3$~~

T.F.D.

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 2n - 1}{n^2 + 5} = 3 \text{ Diverges.}$$

2.0.4 Example:

$$\sum_{n=1}^{\infty} \frac{\tan^{-1} n}{n^2}$$

Compare to $\sum \frac{\frac{\pi}{2}}{n^2} = \frac{\pi}{2} \sum \frac{1}{n^2} \leftarrow \text{Converge}$

$$\sum \frac{\tan^{-1} n}{n^2} < \sum \frac{\frac{\pi}{2}}{n^2}$$

Converge

$\nearrow$
smaller numerator
smaller number.

2.1 More stuff with the comparison test

Do these series converge or diverge?

$$1 + 3 + 5 + \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \text{Diverge}$$

$$1 + 5 + 3 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \text{Converge}$$

$$.01 + .001 + .0001 + \dots + .000000001 + \sum_{n=2}^{\infty} \left(\frac{3}{2}\right)^n \mathcal{D}$$

Fact: You can “chop off” any finite number of terms from the start of a series, and that won’t affect convergence/divergence.

2.1.1 Teaching example:

Compare these two series:

$$\begin{aligned} \sum \frac{1}{2^n} &= \frac{1}{1} + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \frac{1}{64} + \dots \mathcal{C} \\ \sum \frac{1}{n!} &= \frac{1}{1} + \frac{1}{1} + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \frac{1}{720} + \dots \mathcal{C} \end{aligned}$$

↑
smaller terms

Restatement of the Comparison Test

Suppose that $\sum a_n$ and $\sum b_n$ are series with positive terms.

- if $\sum a_n$ converges and $b_n \leq a_n$ for all $n \geq N$, then $\sum b_n$ also converges.
- if $\sum a_n$ diverges and $b_n \geq a_n$ for all $n \geq N$, then $\sum b_n$ also diverges.